# Module 6: Day 1 Command Reference Card

Quick reference for all command types and options covered today.

## General Options (Available on All Commands)

| Option | Type | Default | Description |
|---|---|---|---|
| `type` | str | (required) | Command type (`shell`, `ssh`, `regex`, etc.) |
| `cmd` | str | (required) | Command content (meaning varies by type) |
| `save` | str | - | Save output to a file path |
| `metadata` | dict | - | Key-value pairs logged but not executed |
| `exit_on_error` | bool | `True` | Stop playbook on non-zero return code |
| `error_if` | regex | - | Raise error if pattern matches output |
| `error_if_not` | regex | - | Raise error if pattern does NOT match output |
| `only_if` | condition | - | Execute only if condition is true |
| `loop_if` | regex | - | Retry if pattern matches output |
| `loop_if_not` | regex | - | Retry if pattern does NOT match output |
| `loop_count` | int | `3` | Max retry iterations |
| `background` | bool | `False` | Run as background process |
| `kill_on_exit` | bool | `True` | Kill background process when playbook ends |

## Command Types

### `shell`:Local Command Execution

```yaml
- type: shell
  cmd: nmap -T4 $TARGET
  # Optional:
  command_shell: /bin/bash      # Default: /bin/sh
  interactive: False            # Interactive mode
  creates_session: "name"       # Create named session
  session: "name"               # Reuse session
  command_timeout: 15           # Timeout for interactive mode
```

### `ssh`:Remote Command Execution

```yaml
- type: ssh
  cmd: id
  hostname: $TARGET
  username: user
  password: user
  # Optional:
  port: 22
  key_filename: /path/to/key
  timeout: 60
  creates_session: "name"
  session: "name"
  interactive: False
  command_timeout: 15
  prompts: ["$ ", "# ", "> "]
  clear_cache: False
```

### `sftp`:File Transfer

```yaml
- type: sftp
  cmd: put                      # or "get"
  local_path: /local/file
  remote_path: /remote/file
  # Optional:
  mode: "755"                   # File permissions (upload only)
  session: "name"               # Reuse SSH session
  hostname: $TARGET             # Or use cached settings
  username: user
  password: user
```

### `regex`:Pattern Matching

```yaml
- type: regex
  cmd: "pattern with (groups)"
  output:
    VARNAME: "$MATCH_0"
  # Optional:
  mode: findall                 # findall, search, split, sub
  input: RESULT_STDOUT          # Variable name (without $)
  replace: "replacement"        # For mode: sub
```

### `debug`:Inspect and Troubleshoot

```yaml
- type: debug
  cmd: "Message with $VARIABLES"
  # Optional:
  varstore: False               # Dump all variables
```

```
  wait_for_key: False          # Pause for Enter
  exit: False                  # Stop playbook
```

### `setvar`:Set a Variable

```
- type: setvar
  variable: VARNAME            # Without $
  cmd: "value with $SUBSTITUTION"
  # Optional:
  encoder: base64-encoder      # base64-encoder/decoder, rot13,
urlencoder/decoder
```

### `mktemp`:Create Temporary File/Directory

```
- type: mktemp
  variable: TEMPFILE           # Without $
  cmd: file                    # "file" or "dir"
```

### `sleep`:Pause Execution

```
- type: sleep
  seconds: 5
  # Optional:
  random: False
  min_sec: 0                   # Lower bound for random sleep
```

### `loop`:Iterate

```
- type: loop
  cmd: "items(LIST_VAR)"      # or "range(0, 10)" or "until($X == done)"
  commands:
    - type: shell
      cmd: echo $LOOP_ITEM    # $LOOP_ITEM for items(), $LOOP_INDEX for
range()
  # Optional:
  break_if: "$DONE == yes"
```

### `include`:Include Another Playbook

```
- type: include
  local_path: path/to/other_playbook.yml
```

## Builtin Variables

| Variable | Set By | Description |
|---|---|---|
| `$RESULT_STDOUT` | Most commands | Output of the last command |
| `$RESULT_RETURNCODE` | Most commands | Return code of the last command |
| `$MATCH_0`, `$MATCH_1`, ... | `regex` | Captured groups from regex |
| `$REGEX_MATCHES_LIST` | `regex` | All matches as a list |
| `$LOOP_ITEM` | `loop` (items) | Current list element |
| `$LOOP_INDEX` | `loop` (range/until) | Current iteration index |

## Conditional Operators

| Operator | Example | Description |
|---|---|---|
| `==` | `$PORT == 80` | Equal |
| `!=` | `$CODE != 0` | Not equal |
| `>`, `>=`, `<`, `<=` | `$COUNT > 1` | Numeric ordering |
| `=~` | `$OUT =~ .*open.*` | Regex match |
| `!~` | `$OUT !~ .*closed.*` | Regex no match |
| `is` | `$VAR is None` | Identity check |
| `is not` | `$VAR is not None` | Negated identity |
| `not` | `not $FLAG` | Negation |